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Summary

This documentation elaborates and addresses the issue of currently there are no direct way to setup the live migration network for PCD compute hosts. In Such cases how to setup a separate live migration network for VMs between computes in pcd?

Solution

The `live_migration_scheme` support `tls`, `ssh` and `tcp` as scheme types

To setup TLS migration scheme

- On every pcd hypervisor host setup the `nova_override.conf` with below option:

```
[libvirt]
live_migration_uri = qemu+tls://%s/system?no_verify=1&pkipath=/etc/pf9/certs/libvirt
live_migration_scheme = tls
live_migration_inbound_addr = 192.168.250.10
live_migration_with_native_tls = true
live_migration_tunnelled = false

cpu_mode = custom
cpu_models = Skylake-Server-IBRS
```

The `live_migration_inbound_addr` should point to the IP address assigned to the host that is to be used for the migration network traffic.

- Restart the the `pf9-ostackhost.service`

```
systemctl restart pf9-ostackhost.service
```

To use ssh or tcp as scheme

- On every pcd hypervisor host setup the `nova_override.conf` with below option:

```
[libvirt]
live_migration_uri = qemu+tls://%s/system?no_verify=1&pkipath=/etc/pf9/certs/libvirt
live_migration_scheme = tcp # OR ssh
live_migration_inbound_addr = 192.168.250.10
live_migration_with_native_tls = false
live_migration_tunnelled = false

cpu_mode = custom
cpu_models = Skylake-Server-IBRS
```

The `live_migration_inbound_addr` should point to the IP address assigned to the host that is to be used for the migration network traffic.

- Restart the the `pf9-ostackhost.service`

Observations/caveats/issues:

- `live_migration_tunneled` should be kept as `false` for most case it would be the default value. With this set as `true` the `qemu` copy option will get tunneled through the `libvirt` connection for encryption option. This comes with performance penalties.
- The TLS scheme offers the best balance of performance and speed for live migration transfers
- TCP can be the fastest but it is unencrypted. Not useful unless it's really the only way to setup.
- SSH is more used in development and testing, not recommended in production as being little less secure than TLS option.

Reference logs

- Upon successful operation, On the target hypervisor following messages should be noted:

```
2026-02-22 16:24:03.941 INFO nova.virt.libvirt.migration [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b
arvind@platform9.com Tenant1] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] Increasing downtime to 50 ms
after 0 sec elapsed time
2026-02-22 16:24:04.904 INFO barbicantclient.base [req-31bbd1ff-337a-48c8-80d8-56264753cd60 nova services]
Calculated Secrets uuid ref: secrets/5af99e50-1421-4cec-b235-b117e94fcb05
2026-02-22 16:24:05.544 INFO nova.compute.manager [req-52621501-02e3-40a2-b566-e70e9f57a725 None None]
[instance: ed387513-4254-4b4e-9e0c-0073ed460711] VM Paused (Lifecycle Event)
2026-02-22 16:24:05.743 INFO nova.virt.libvirt.driver [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b
arvind@platform9.com Tenant1] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] Migration running for 1 secs,
memory 100% remaining (bytes processed=0, remaining=0, total=0); disk 100% remaining (bytes processed=0,
remaining=0, total=0).
2026-02-22 16:24:06.248 INFO nova.virt.libvirt.driver [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b
arvind@platform9.com Tenant1] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] Migration operation has
completed

2026-02-22 16:24:06.249 INFO nova.compute.manager [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b
arvind@platform9.com Tenant1] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] _post_live_migration() is
started..
2026-02-22 16:24:08.046 WARNING nova.compute.manager [req-4508ef35-1f95-4286-9bb3-96990ea72484 nova services]
[instance: ed387513-4254-4b4e-9e0c-0073ed460711] Received unexpected event network-vif-plugged-5038fec4-1122-
435c-b1a9-4105e39c5b52 for instance with vm_state active and task_state migrating.
2026-02-22 16:24:08.047 WARNING nova.compute.manager [req-4508ef35-1f95-4286-9bb3-96990ea72484 nova services]
[instance: ed387513-4254-4b4e-9e0c-0073ed460711] Received unexpected event network-vif-plugged-5038fec4-1122-
435c-b1a9-4105e39c5b52 for instance with vm_state active and task_state migrating.
2026-02-22 16:24:10.545 WARNING nova.compute.manager [req-b934accb-8c51-495c-a2d6-21d3da8c59a3 nova services]
[instance: ed387513-4254-4b4e-9e0c-0073ed460711] Received unexpected event network-vif-plugged-5038fec4-1122-
435c-b1a9-4105e39c5b52 for instance with vm_state active and task_state migrating.
2026-02-22 16:24:12.571 WARNING nova.compute.manager [req-eeclf5fd-dff7-43a5-90ed-83809c645c71 nova services]
[instance: ed387513-4254-4b4e-9e0c-0073ed460711] Received unexpected event network-vif-plugged-5038fec4-1122-
435c-b1a9-4105e39c5b52 for instance with vm_state active and task_state migrating.
2026-02-22 16:24:13.987 INFO os_vif [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b arvind@platform9.com Tenant1]
Successfully unplugged vif VIFOpenVSwitch(active=True,address=fa:16:3e:6e:6f:56,bridge_name='br-
int',has_traffic_filtering=True,id=5038fec4-1122-435c-b1a9-4105e39c5b52,network=Network(92525eac-c837-48ff-
912a-
1e6109b00c72),plugin='ovs',port_profile=VIFPortProfileOpenVSwitch,preserve_on_delete=False,vif_name='tap5038f
ec4-11')
2026-02-22 16:24:13.989 INFO nova.virt.libvirt.driver [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b
arvind@platform9.com Tenant1] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] Deleting instance files
/opt/data/instances/ed387513-4254-4b4e-9e0c-0073ed460711_del
2026-02-22 16:24:13.990 INFO nova.virt.libvirt.driver [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b
```

```
arvind@platform9.com Tenant1] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] Deletion of  
/opt/data/instances/ed387513-4254-4b4e-9e0c-0073ed460711_del complete  
2026-02-22 16:24:21.303 INFO nova.compute.manager [-] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] VM  
Stopped (Lifecycle Event)  
2026-02-22 16:24:32.739 INFO nova.compute.manager [req-68bd38c8-80b7-44b4-ae3-474b1e6eb20b  
arvind@platform9.com Tenant1] [instance: ed387513-4254-4b4e-9e0c-0073ed460711] Migrating instance to  
971596f1-50f1-4406-b569-3fe31d69702c finished successfully.
```